Debiased Learning of Self-Labeled Twitter Data for User Demographic Prediction

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***Abstract*—Labeling sufficient data for supervised learning remains an open challenge in social network analysis. An alternative is to collect self-labeled data, i.e. the data labeled by their owners. Emmery et al show that standard models can be trained and perform well on self-labeled data, suggesting the effectiveness of this approach. In this paper, we argue self-labeled data may not be representative of the population. Taking Twitter demographic prediction as an example, we show the popular FastText model standardly trained on self-labeled data does not generalize well on random testing samples. We then present a new learner DeFastText that aims to correct data bias using the kernel means matching technique. In experiment, we show it achieves lower generalization errors than FastText. This research raises an attention of the data bias problem when learning from self-labeled data in social network analysis.**

***Index Terms*—self-labeled data, demographic prediction, fasttext, kernel means matching, concept drift**

I. Introduction

Labeling sufficient data for supervised learning remains an open challenge in social network analysis. Indeed, going over a user's tweets to annotate his political bias is often an inefficient and tedious work, whether it is done by dedicated annotators or crowdsourced to the public.

An emerging alternative is to collect *self-labeled data*, i.e., data that are labeled by their owners, often in an indirect way. Figure 1 shows an example on Twitter. The first tweet 'I'm a man' implies the gender attribute of its poster, while the second tweet 'Honestly, I'm black...' implies the race attribute of its poster. When working on tweet-based gender or race prediction, these tweets can be considered as self-labeled data.

Emmery et al [6] is, to our knowledge, the first work that empirically evaluates the effectiveness of self-labeled twitter data on tweet-based gender prediction. Training the popular FastText prediction model [5] on self-labeled tweets, they show the model achieves approximately 70% accuracy on testing sets, which also consist of self-labeled data. Their study points to a promising land of using self-labeled data for supervised learning in social network analysis.

Fig. 1. Two Example Self-Labeled Tweets.

In this paper, we argue that self-labeled data may not be representative of the population, where a trained model is often expected to generalize on. To illustrate this point, we crawled both self-labeled tweets listed in [6] and random tweets listed in [4], and compare their country distribution in Figure 2. We see that self-labeled users mostly come from countries 1, 5, 6, 12 and 15, whereas random users mostly come from countries 1, 2, 4, 5 and 6. There is a clear gap between self-labeled and random distributions, possibly to different degrees of self-awareness in different cultures. However, such gap is ignored in the prior study.

Picking on Emmery et al's example of Twitter gender prediction, we show that FastText standardly learned from self-labeled data does not generalize well on random testing sets, which we crawled from Twitter and consist of manually-labeled data. To mitigate such bias, we employ the sampling bias correction technique when learning from self-reported data sets. Specifically, we propose a novel Debiased FastText (**DeFastText**), which integrates FastText and Kernel Means Matching (KMM) [3]. DeFastText aims to train FastText in an embedded space where self-labeled data and random data have similar distributions. We jointly optimize the FastText classifier and the KMM correction weights. In experiment, we show the proposed learner allows FastText to achieve lower

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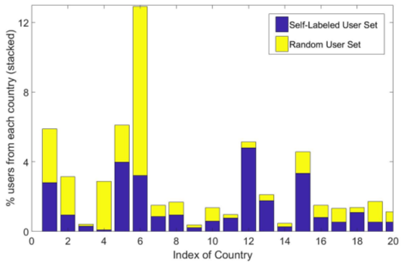

Fig. 2. Distribution of Users across Countries

error on random testing sets, compared with the standard learner when trained on self-labeled data. This supports our argument that self-labeled data need correction when used in supervised learning.

II. Methodology

A. Problem Setting

Let $(x_1, y_1), \ldots, (x_n, y_n)$ be n self-labeled instances, where $x_i \in \mathbb{R}^p$ is feature vector of the i_{th} instance and y_i is its self-annotated label (e.g., a demographic feature like gender). Let $t_1, \ldots, t_m$ be a set of m random instances, where t_i is feature vector of the i_{th} instance. Our task is to train a prediction model f from the self-labeled set, and apply it to predict labels of the random set. A technical challenge is that x_i's and t_i's are generated from different distributions.

B. A Revisit of FastText

For any user x, its prediction with FastText model is defined as $f(x) = [\hat{y}_1(x), \ldots, \hat{y}_c(x)]^T$, where c indicates there are c demographic classes, and $\hat{y}_j(x)$ is the predicted probability that x belongs to the j_{th} class. Specifically, $\hat{y}_j(x)$ is defined as

$$\hat{y}_j(x) = \frac{\exp(B_{j:}Ax)}{\sum_{r=1}^{c} \exp(B_{r:}Ax)} \tag{1}$$

where $A \in \mathbb{R}^{k \times p}$ is a feature mapping matrix, $B \in \mathbb{R}^{c \times k}$ is a classifier matrix and $B_{j:}$ denotes its j_{th} row. Both A, B will be learned by minimizing the entropy loss

$$L(f) = -\frac{1}{n}\sum_{i=1}^{n} y_i^T \log f(x_i) \tag{2}$$

where the logarithm function is element-wise.

C. DeFastText

To correct data bias when learning FastText, we employ the kernel means matching (KMM) [2] technique, and present a new learner called Debiased FastText (DeFastText).

KMM aims to reduce distribution gap between training data and testing data by reweighing the former. Let ϕ be a mapping from raw feature space to a higher dimensional space, and let β_i be the weight for instance x_i. Kernel Means Matching (KMM) aims to reduce distribution gap between training data and testing data by reweighing the former. KMM learns β_i's by minimizing

$$D(x,t) = \left|\left| \frac{1}{n}\sum_{i=1}^{n} \beta_i \phi(x_i) - \frac{1}{m}\sum_{i=1}^{m} \phi(t_i) \right|\right|^2 \tag{3}$$

A naive way to integrate KMM and FastText, is to first learn β_i's through (3) and then learn a FastText model by minimizing a weighted entropy loss, e.g., $L_\beta(f) = -\frac{1}{n}\sum_{i=1}^{n} \beta_i y_i^T \log f(x_i)$. However, this may suffer from sub-optimal performance because the weights and the classifier are not learned in the same space – weights are learned in the raw space whereas the classifier B is learned in the embedded space of A.

To this end, we propose to jointly learn weights and classifier, both in the embedded space of A, by solving the following bi-level optimization problem

$$\begin{aligned} &\min_{A,B} \; -\frac{1}{n}\sum_{i=1}^{n} \hat{\beta}_i \, y_i^T \, \log f(x_i), \\ &s.t. \; \{\hat{\beta}_i\} = \arg\min_{\{\beta_i\}} \left|\left| \frac{1}{n}\sum_{i=1}^{n} \beta_i \phi(Ax_i) - \frac{1}{m}\sum_{i=1}^{m} \phi(At_i) \right|\right|^2 \end{aligned} \tag{4}$$

Solving (4) is non-trivial, because the lower-level optimization task includes a parameter A that needs to be optimized at the higher-level task. We propose an efficient solver that alternately optimizes model (A, B) and weight β_i's using stochastic gradient descent.

D. Optimization of DeFastText

Let I_s be an index set of the instances in one update iteration. The unnormalized entropy loss on I_s is

$$L_s = -\sum_{i \in I_s} \hat{\beta}_i \, y_i^T \, \log f(x_i) = \sum_{i \in I_s} L_{si} \tag{5}$$

where we introduce

$$L_{si} = -\hat{\beta}_i \, y_i^T \, \log f(x_i) = -\hat{\beta}_i \sum_{j=1}^{c} y_{ij} \log(\hat{y}_{ij}) \tag{6}$$

and

$$\hat{y}_{ij} = f(x_i)_j = \frac{e^{B_{j:}Ax_i}}{\sum_{r=1}^{c} e^{B_{r:}Ax_r}} \tag{7}$$

The model parameters are A, B and β. In the following, we propose to optimize them alternately.

(i) Update B. Fixing A and β, we update the j_{th} row of B by

$$B_{j:} = B_{j:} - \gamma \cdot \frac{\partial L_s}{\partial B_{j:}} = B_{j:} - \gamma \cdot \sum_{i \in I_s} \beta_i (\hat{y}_{ij} - y_{ij})(Ax_i)^T \tag{8}$$

where γ is learning rate.

(ii) Update A. Fixing B and β, we update the i_{th} row of A by

$$A_{j:} = A_{j:} - \gamma \cdot \frac{\partial L_s}{\partial A_{j:}} = A_{j:} - \gamma \cdot \sum_{i \in I_s} \beta_i (\hat{y}_i - y_i)^T B_{:j} x_i^T \tag{9}$$

(iii) Update β. Fixing A and B, we optimize β by minimizing $D_A(x,t)$ with proper regularization on its complexity, i.e.,

$$\begin{aligned} &\min_{\beta} \; D_A(x,t), \\ &s.t. \; \forall i, \; \beta_i \in [0, B_d] \; \text{ and } \; \left| \frac{1}{n}\sum_{i=1}^{n} \beta_i - 1 \right| \leq \epsilon \end{aligned} \tag{10}$$

TABLE I
CLASSIFICATION PERFORMANCE OF EXAMINED ALGORITHMS ON GENDER AND RACE PREDICTION TASKS.

	Metrics	FastText	RoFastText
Gender Prediction	Testing Error on Self-Labeled Users	**0.3326**±0.004	0.3406±0.012
	Testing Error on Random Users	0.4726±0.022	**0.4415**±0.0336
Race Prediction	Testing Error on Self-Labeled Users	0.4589±0.008	**0.4557**±0.008
	Testing Error on Random Users	0.4724±0.035	**0.4569**±0.015

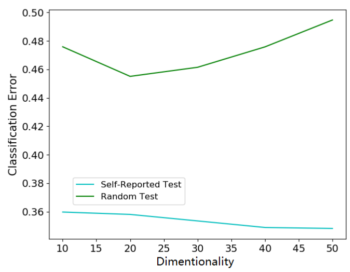

Fig. 3. Generalization Error of DeFastText versus Dimension of Latent Space.

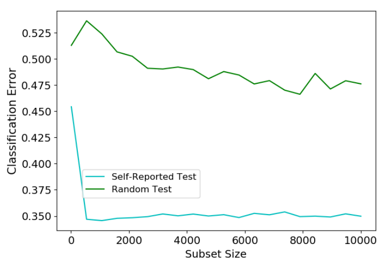

Fig. 4. Generalization Error of DeFastText versus Number of Self-Labeled Users in the Training Set.

where B_d and ϵ are hyper-parameters. By standard arguments, we can easily show that $D_A(x,t)$ can be written as

$$D_A(x,t) = \frac{1}{n^2}\beta^T K_{xx}\beta - \frac{2}{mn}\beta^T K_{xt}1 + \frac{1}{m^2}1^T K_{tt}1, \quad (11)$$

where $\beta = [\beta_1, \ldots, \beta_n]^T$, 1 is a properly-sized vector of ones, $K_{xx} \in \mathbb{R}^{n\times n}$, $K_{tt} \in \mathbb{R}^{m\times m}$ and $K_{xt} \in \mathbb{R}^{n\times m}$ are Gram matrices defined as

$$K_{xx}(i,j) = k(Ax_i, Ax_j), \quad (12)$$

$$K_{tt}(i,j) = k(At_i, At_j), \quad (13)$$

$$K_{xt}(i,j) = k(Ax_i, At_j). \quad (14)$$

Therefore, $D_A(x,t)$ is a quadratic function of β and (10) can be solved by any standard quadratic optimization tool.

III. EXPERIMENT

We examine our hypothesis and the proposed algorithm on four publicly listed Twitter user sets: Self-Labeled Twitter User Gender Data Set [6], Random Twitter User Gender Data Set [4], Self-Labeled Twitter User Race Data Set [7] and Random Twitter User Race Data Set [1].

We examined two prediction tasks: gender prediction and race prediction. For each algorithm, we ran it over 20 random trials and reported the average performance. We initialize the learning rate for FastText as 0.01 and set the dimension of latent space k to 30 (same as the previous study). We chose rbf kernel with default hyperparameter.

Testing results of both algorithms are shown in Table I. We observe that FastText does not generalize as well on random user as self-labeled user. Comparatively, DeFastText achieves smaller error on the random testing set, suggesting its effectiveness in correcting data bias and supporting our hypothesis on the bias of self-labeled data. In Figure 3 and 4, we show the generalization performance of DeFastText versus the number of self-labeled users and dimension of latent space respectively. We see that generalization error is significantly reduced as more self-labeled instances are used in training and the proposed DeFastText is comparatively robust to the latent space dimension.

IV. CONCLUSION

In this paper, we initiate an investigation on the data bias problem when learning from self-labeled data in social network analysis. We argue self-labeled data may not be representative of the population, and empirically show that FastText standardly trained on self-labeled data does not generalize well on random data in the tweet-based user demographic prediction task. We present DeFastText, a new learner that corrects the bias using kernel means matching, and show it achieves lower generalization error than FastText.

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